Problem 18

Let

$$D = \left\{ \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix} : |z| = 1 \right\} \subset SU(2), \qquad N = N_{SU(2)}(D).$$

You may use the results of Problem 11 in HW02 and Problem 17 in HW03. Identify $N/D \cong \mathbb{Z}_2$, and let $\pi : N \to \mathbb{Z}_2$ be the quotient homomorphism. Show that the short exact sequence

$$1 \longrightarrow D \xrightarrow{\iota} N \xrightarrow{\pi} \mathbb{Z}_2 \longrightarrow 1,$$

where ι is the inclusion, does not split. Equivalently, show that there is no group homomorphism $s : \mathbb{Z}_2 \to N$ satisfying $\pi \circ s = \mathrm{id}_{\mathbb{Z}_2}$.

Problem 19

Consider the two representations of $SU(2)$ on $\mathbb{C}^2$ given by

$$\rho(u) = u, \qquad \overline{\rho}(u) = \overline{u},$$

where the bar denotes entrywise complex conjugation. Show that they are equivalent by finding a fixed matrix $S \in GL(2, \mathbb{C})$ such that

$$\overline{u} = SuS^{-1} \qquad \forall u \in SU(2).$$

Problem 20

For each of the following groups, show that the defining representation $\rho(u) = u$ and the conjugate representation $\overline{\rho}(u) = \overline{u}$ are inequivalent over $\mathbb{C}$.

(1) $U(2)$ acting on $\mathbb{C}^2$.

(2) $SU(N)$ acting on $\mathbb{C}^N$, for $N > 2$.

Hint. Consider scalar matrices in the center of each group.